CSCI 441 - Software Engineering

SuperAsk.AI: AI-Powered Government Service Assistant

Report One, Part One

Team AO3

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Project URL: https://github.com/Monunhh/Super-ask.git

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Individual Contributions Breakdown

Team Member	Contribution to Report One, Part One
Yean Fong	Wrote and organized Section 2: Goals, System Requirements, and Analysis; prepared Section 3: Project Management, Gantt chart, team responsibilities, and references; captured prototype screenshots and integrated/formatted the final report.
Doungkhe Neang	Contributed to writing Section 1: decomposition into sub-problems, glossary, and government-service content research.
Sereyprophea Sarin	Contributed to writing and reviewing Section 1: Customer Problem Statement; assisted with documentation, editing, requirement review, and QA-related content.
Phallamonunhh Chun	Contributed to the technical prototype, including AI/RAG-related functionality and technical input supporting the system requirements.
Bunkheang Heng	Contributed to development of the technical prototype, including web/API, system architecture, database, and integration-related work.
Heejae Choi	Contributed to UX/UI design and development of the prototype interface used to illustrate the User Interface Requirements.

1. Customer Problem Statement

Problem Statement

Scenario 1: High School Student Graduate

Right after completing my BACII National Examination, I wanted to move quickly and apply for university. I went to the application site to check the required documents, applying to both national and international universities at once. Across every application, the same two documents kept coming up: my high school diploma and my BACII results. For the results, I used the cropped version posted by the Ministry of Education on Facebook, since that was the only version available to me at the time.

After I submitted my application, I was told the documents were incorrect and that I needed to contact the ministry for verification. That instruction alone created a new problem, since I had no idea which ministry to go to. There are multiple sub-branches, each with a different location, and nothing on the application told me which one was responsible for verification.

I considered using an agency to handle the process for me, but backed out after hearing from friends that the full service costs $500, with 50 percent required upfront as a deposit before I even receive an acceptance letter. On top of the cost, I was also genuinely intimidated by the idea of navigating a government building on my own.

So I went to my teachers at the public school for guidance. They told me to go to my khan, but couldn't tell me where the office was or which specific one I needed. I then went to a general government office to ask directly, and they referred me to several different locations spread across the city. I followed their instructions one by one, and by the time I reached what was supposedly the correct office, I still didn't have the right documents with me. That entire day was lost.

Once I finally gathered the correct documents and returned, I learned about an additional fee and a late fee, and that my diploma also needed to be officially notarized before it would be accepted. Altogether, the process required layered approval from multiple separate offices and ministries, each with its own deadline and its own late fee structure, none of which were listed together anywhere. The combined waiting time across all these steps ended up being so long that I missed my application deadline entirely, despite starting the process right after my exam results came out.

The core problem throughout was that the information I needed was never in one place. It was scattered across different ministry websites, social media posts, and word of mouth, with no single source telling me the required documents, the correct offices, the fees, or the deadlines all at once.

Scenario 2: First-time Mothers

After giving birth to my son, I spent a week resting in the hospital. Before discharge, the doctors handed me a detailed medical report on my son and told me to get him registered. That was the extent of the guidance; I wasn't told which ministry or office was actually responsible for processing it.

Not wanting to spend hours checking every possible ministry, I decided to set aside time and figure it out myself. But time was already tight: I was on pregnancy leave, though still expected to do remote work as an assistant to my team; I was physically recovering from labour, and I was hosting daily visits from family coming to see the new baby. Realistically, I could only carve out one full day to handle the registration in person.

Weekends were my only free window, so that's when I tried to go. Each time, something didn't line up: the office's shift had ended, it was their lunch break, or it simply wasn't a working day for them. My free time and their available hours never overlapped. Between recovering from childbirth, caring for the baby alone, and managing remote work, I couldn't shift my schedule to match theirs, and neither could they shift to match mine. Two weeks passed this way, with every attempt missing.

A month in, I finally got a Monday off, a weekday, when the office would actually be open. I brought the documents the hospital gave me and went to register. Only then did I learn several things at once: I had already missed the registration deadline, incurring late fees on top of the original fee; the hospital's report wasn't sufficient on its own, since I needed additional specific documents; and I was at the wrong office entirely, since I needed to go through my khan, not the central ministry.

So a month of effort, plus money spent on the fee and now a late penalty, resulted in nothing being filed. And I still don't know exactly which documents I need to bring or how they need to be formatted so that next time, I can submit everything correctly in one visit, without having to rely on informally checking Telegram and Facebook updates from both the ministry and the khan just to stay current, on top of whatever unspoken procedural rules aren't posted anywhere at all.

Scenario 3: National Election Registration

Election season came around, and I knew I needed to register to vote. The problem was, the deadline wasn't sent to me directly. It was only announced through the Ministry's Telegram channel and Facebook page. So I had to go check for myself, scrolling through weeks of posts to find the actual date, because the feed was flooded with unrelated announcements, reposts, and comments, and I couldn't always tell which post was the most recent or which one had been quietly updated or corrected later. I bookmarked what I thought was the right date, but I had no way of confirming whether it was still accurate without checking again and again.

On top of that, I wasn't sure if my ID was even valid for registration anymore. Nobody tells you outright; you find out the procedure depends on your age group. Younger first-time voters go through one process, people renewing an expired ID go through another, and everyone assumes you already know which category you fall into. I didn't.

Here's the part that confused me most: registering to vote itself is free. But if my ID happens to be outdated, I get charged a separate fee to renew a fee that has nothing to do with the election, but that I have to pay anyway just to be eligible. And if I miss that ID renewal window, I don't just pay a penalty; I lose the ability to vote entirely, with no appeal and no second window. So a paperwork issue that's

technically unrelated to the election becomes the single point of failure that decides whether I get a say at all.

I ended up spending more time trying to verify whether my information was current than actually completing the registration.

Scenario 4: Government Officers

From the other side of the counter, the picture looks different, but the outcome is the same. The office is short-staffed, and the people who are there are working shifts because they're human beings who also need lunch breaks, rest, and time off. When a citizen shows up expecting help exactly when the office happens to be short-handed or between shifts, there's often no one immediately free to assist, not because no one cares, but because there simply aren't enough people to cover every hour without gaps.

Meanwhile, the staff who are on duty are constantly interrupted. Someone walks in with a question, another person cuts the line, and a third shows up with the wrong documents entirely. Each interruption breaks whatever task the officer was in the middle of completing. Multiply that by dozens of people a day, and very little progress is made on any single case.

A lot of the workload is also self-inflicted by the process itself: citizens frequently arrive with documents that aren't formatted the way the system requires, missing the specific keywords or fields needed for official filing. The officer then has to explain what's wrong, send the person away to fix it, and wait for them to come back. Sometimes days or weeks later, restarting the same conversation from scratch. This repetitive back-and-forth eats up time that could otherwise go toward processing new cases.

And because the office doesn't have the staffing to individually track every applicant's status, there's no consistent way to flag "this person is close to a deadline" or "this person already tried once and got turned away." Each visit is treated as if it's the first, because there isn't the manpower to do otherwise. It would be a miracle if the citizens were able to access a sample and know what is required of them beforehand so when they come, the document can be registered right away, saving us both the overhead.

Decomposition into Sub-problems

High school graduates

- Identify the procedure for getting school documents notarized or certified.
- Determine what documents are required, such as the high school diploma and BACII results.
- Explain where students need to go and the steps they need to follow.
- Help students understand what documents are accepted when official results or certificates are not yet available.

First-time parents

- Explain how to register a newborn and obtain a birth certificate.
- Identify the required documents.
- Provide information about registration deadlines, locations, and possible fees.

Citizens registering for elections

- Provide information about voter registration deadlines.
- Explain what documents are required and whether they need to be renewed.
- Identify where citizens need to register and whether there are any fees.

Government officers

- Reduce repetitive questions and tasks handled by government staff.
- Organize commonly requested information so citizens can access it themselves.
- Reduce workload and make the process of providing government information more efficient.

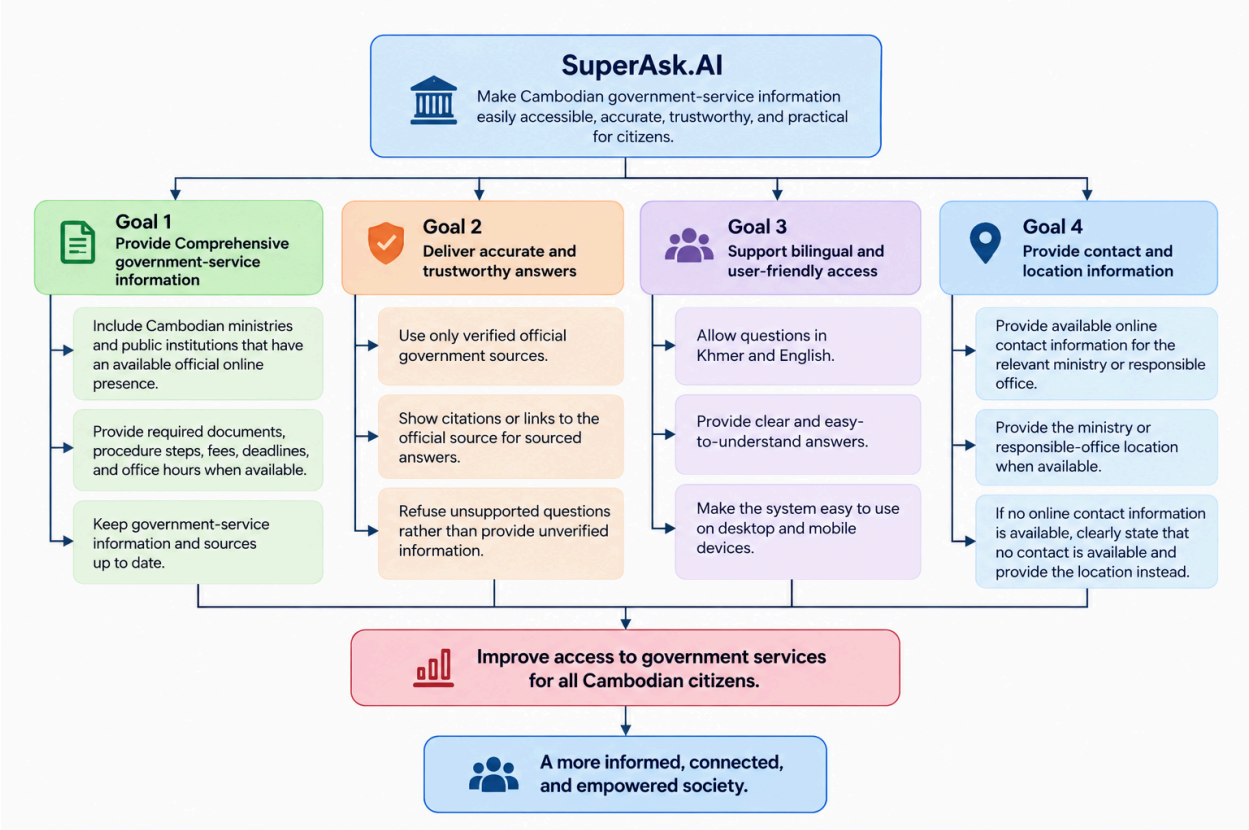
Glossary of Terms

Terms	Meaning
BacII	Cambodia's National Exam.
Notarize	To officially verify that a document or signature is real.
Government Contact Referral	The process of directing a citizen to the relevant government ministry or office by providing available official contact information and location when additional assistance from the government office is needed.
Unsupported Query	A question that SuperAsk cannot answer because there is no approved government information available in its database.
Effective Date	The date when a law, rule, procedure, or requirement officially begins to apply.
Last-Verified Date	The most recent date when information in SuperAsk was checked and confirmed to still be accurate.
Legal Instrument	An official legal document, such as a law, sub-decree, proclamation, regulation, or directive, that provides authority for a government procedure.
Government Procedure	The official steps a citizen must follow to access or complete a government service.
Khan	Administrative district in Cambodia

2. Goals, System Requirements, and Analysis

Goals

The business goals below show how SuperAsk.AI's main objective is divided into four supporting goals: comprehensive government-service information, trustworthy answers, bilingual accessibility, and ministry contact and location information.



Enumerated Functional Requirements

Priority Weight (PW) uses a scale from 1 to 5, where 5 represents the highest-priority requirements and 1 represents the lowest-priority or optional requirements.

Identifier	PW	Requirement
REQ-1	5	The system shall provide a chat interface where users can submit questions about Cambodian government services.
REQ-2	5	The system shall accept questions in Khmer and English.
REQ-3	4	The system should support mixed Khmer-English questions.

REQ-4	5	The system shall retrieve relevant information from verified official government online sources before generating an answer.
REQ-5	5	The system shall generate answers using only relevant information retrieved from approved government sources.
REQ-6	5	The system shall display the official source or citation supporting each sourced answer.
REQ-7	5	The system shall decline to provide an unsupported answer when no verified official source can be found.
REQ-8	5	The system shall provide government-service information for Cambodian ministries and public institutions that have an available official online presence.
REQ-9	4	The system shall allow users to browse or select commonly requested government procedures.
REQ-10	3	The system shall display recent questions or conversations to help users return to previously viewed topics.
REQ-11	4	The system shall provide available online contact information for the relevant ministry or responsible office when such information is available from an official source.
REQ-12	4	The system shall provide the location of the relevant ministry or responsible office when location information is available from an official source.
REQ-13	4	If no online contact information is available, the system shall inform the user that no online contact is available and provide the ministry or responsible-office location instead.
REQ-14	4	The system shall maintain conversation context so users can ask follow-up questions during the same session.
REQ-15	3	The system should support access through the web, Telegram, and Messenger.
REQ-16	5	The system shall provide the required documents for a government procedure when the information is available from a verified official source.
REQ-17	5	The system shall provide the steps required to complete a government procedure in the correct order when available from a verified official source.

REQ-18	5	The system shall provide official fees and applicable late fees when available from a verified official source.
REQ-19	5	The system shall provide applicable deadlines or validity periods when available from a verified official source.
REQ-20	4	The system shall provide available office hours and other relevant office information when provided by a verified official source.

Enumerated Nonfunctional Requirements

The numbering continues from the functional requirements.

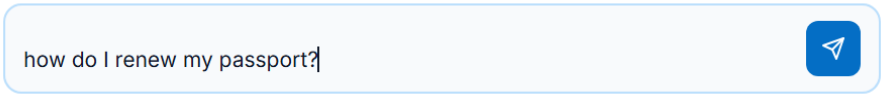
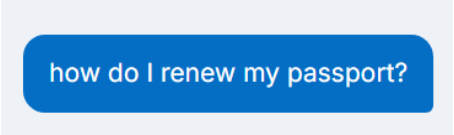
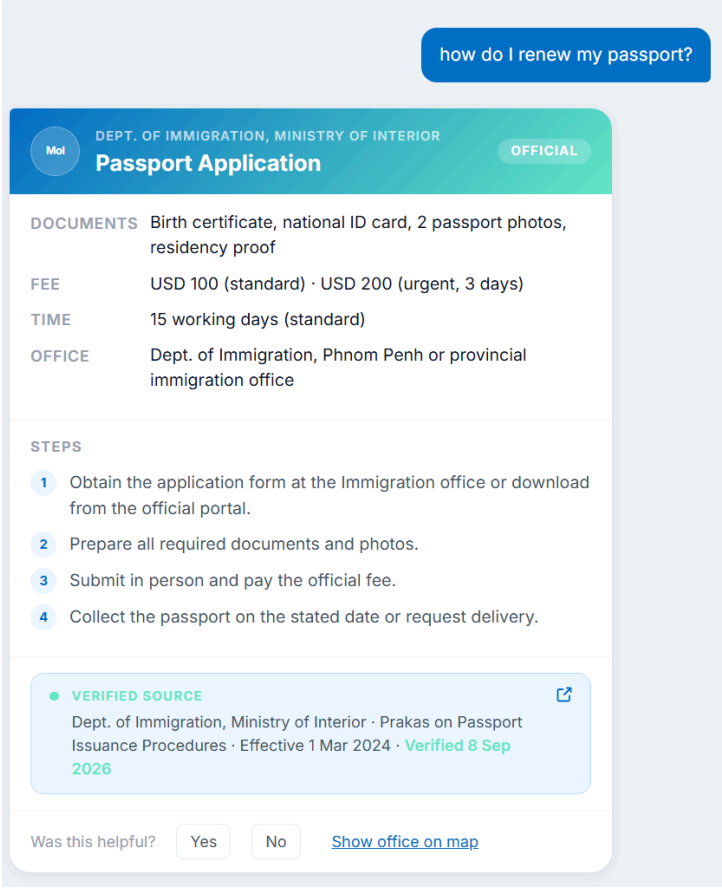
Identifier	PW	Requirement
REQ-21	5	At least **90% of test users** should be able to submit a government-service question through the main chat interface without assistance.
REQ-22	5	During acceptance testing, no sourced answer shall contain a factual claim that is unsupported by its displayed verified official source.
REQ-23	5	The system shall correctly refuse at least **95% of unsupported or out-of-scope questions** included in the project test set.
REQ-24	4	Under normal testing conditions, the system should begin displaying a response within **5 seconds for at least 90% of test queries**.
REQ-25	4	The web interface shall remain usable without horizontal scrolling on screens from approximately **360 pixels wide** through standard desktop resolutions.
REQ-26	4	Khmer and English text shall display without broken, missing, or unreadable characters in the browsers used for project testing.
REQ-27	5	API keys, model credentials, and other system secrets shall remain server-side and shall not be accessible through the citizen-facing browser interface.
REQ-28	4	Every sourced answer shall visually separate the answer content from its official source or citation information.
REQ-29	4	Authorized government content and ministry information shall be updateable through stored data or configuration without requiring changes to the core answer-generation logic.

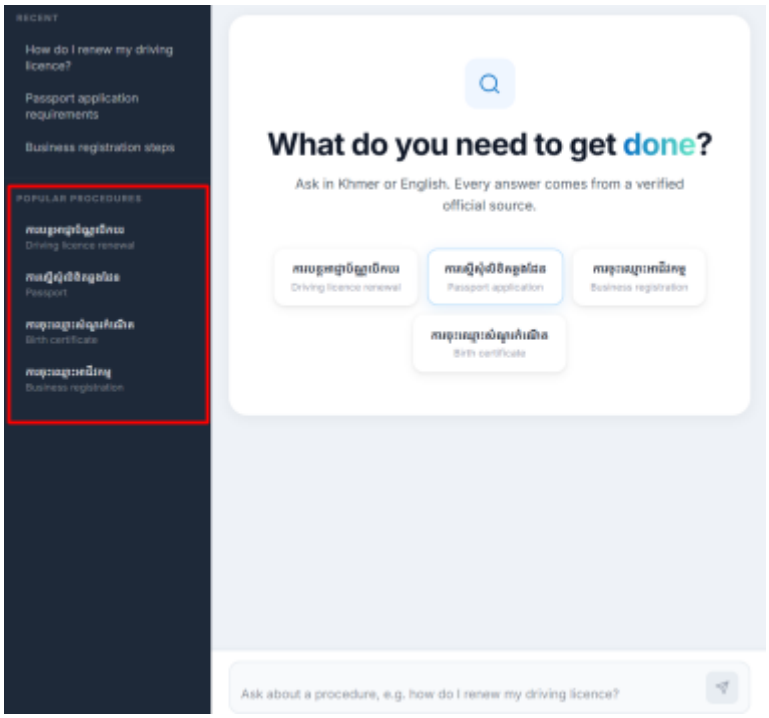
REQ-30	4	Additional ministries and public institutions shall be addable without redesigning the core retrieval and answer-generation architecture.

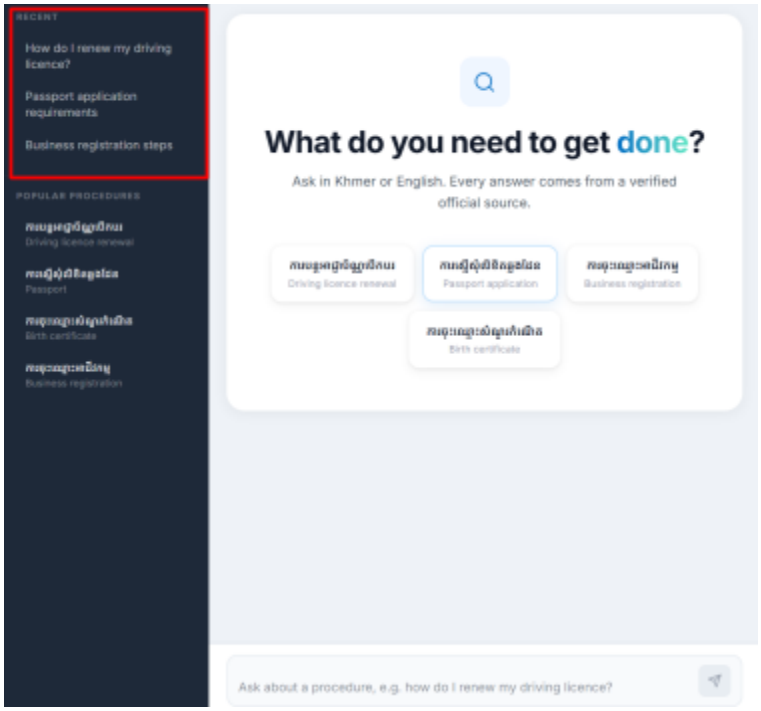
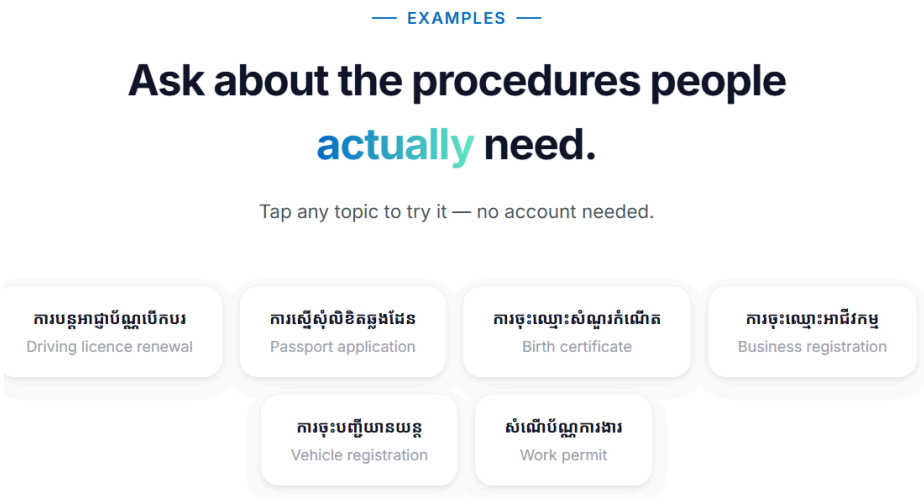
FURPS+ Table

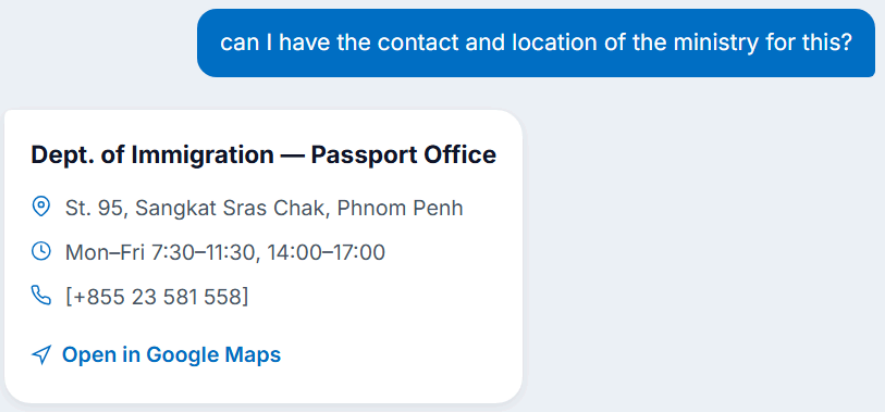
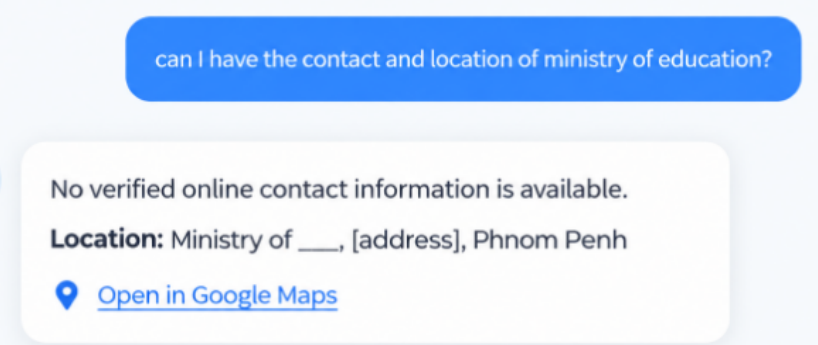
FURPS+ Category	SuperAsk.AI
Functionality	The system supports government-service questions, Khmer and English interaction, verified-source retrieval, citations, procedure steps, required documents, fees, deadlines, office information, ministry contact details, and ministry locations.
Usability	The chat interface should be simple, clearly organized, responsive, and usable on both mobile and desktop devices. At least 90% of test users should be able to submit a question without assistance.
Reliability	The system should answer using verified official sources and refuse unsupported questions rather than guess. Sourced answers must remain consistent with their displayed sources during acceptance testing.
Performance	Under normal testing conditions, the system should begin displaying a response within 5 seconds for at least 90% of test queries.
Supportability	Government information, ministry contacts, office locations, and official sources should be updateable without changing the core answer-generation logic. Additional ministries should also be addable without major architectural redesign.
Security (+)	API credentials, model keys, and other system secrets must remain server-side and inaccessible through the citizen-facing browser interface.
Design / Implementation Constraints (+)	SuperAsk.AI shall operate as a web-based service, use verified official government information for sourced answers, and support responsive access on both mobile and desktop devices.

User Interface Requirements

Identifier	Priority	Requirement
REQ-31	5	A user shall be able to enter a government-service question in the chat input field. 
REQ-32	5	A user shall be able to submit a question to SuperAsk.AI. 
REQ-33	5	A user shall be able to view the answer generated by SuperAsk.AI. 
REQ-34	5	A user shall be able to view the verified official source or citation supporting the answer.

		VERIFIED SOURCE Dept. of Immigration, Ministry of Interior · Prakas on Passport Issuance Procedures · Effective 1 Mar 2024 · Verified 8 Sep 2026
REQ-35	4	A user shall be able to select a popular procedure from the interface. 

REQ-36	4	A user shall be able to view recent questions or conversations. 
REQ-37	4	A user shall be able to use the interface in Khmer or English. 
REQ-38	4	A user shall be able to view available ministry contact and location information when provided by an official source.

		
REQ-39	4	If no online contact information is available, the interface shall state that no contact is available and display the ministry or responsible-office location. 

3. Project Management

Plan of Work

Following the submission of Report One, our team will continue developing the system through two main iterations. The first iteration will focus on developing and integrating the core functions needed for the first demonstration, including the web interface, bilingual question handling, retrieval from verified government sources, citation display, and guardrails for unsupported questions. The second iteration will focus on refinement, additional ministry coverage, testing, and improvements based on feedback from the first demonstration. The short-term activities are described in greater detail because they are the immediate priorities of the team. Later milestones may be adjusted based on technical progress, testing results, and instructor feedback.

Period	Planned Work	Milestone
Sep 21 - 25, 2026	Complete customer requirements, business goals, functional/nonfunctional requirements, UI requirements, project management, and preliminary interface designs.	Report #1 Parts 1 & 2

Sep 26 - 30, 2026	Review and revise requirements; ensure consistency between the problem statement, requirements, prototype, and team responsibilities.	Full Report #1 – Sep 30
Oct 1 - 6, 2026	Continue collecting and verifying official government information. Develop interaction diagrams and begin implementing the retrieval and answer-generation workflow.	Report #2 Part 1 – Oct 6
Oct 7 - 12, 2026	Develop class/system architecture diagrams. Continue development of the web interface, database, retrieval system, and bilingual processing.	Report #2 Part 2 – Oct 12
Oct 13 - 21, 2026	Integrate the interface, retrieval pipeline, citation system, guardrails, and verified-source database. Conduct initial integration testing.	Full Report #2 – Oct 21
Oct 22 - Nov 1, 2026	Implement the core use cases selected for the first demonstration. Test Khmer and English queries and verify responses against official sources.	First-demo implementation
Nov 2–8, 2026	Debug the integrated system, test unsupported/out-of-scope questions, improve response quality, and prepare the demonstration.	First-demo preparation
Nov 9, 2026	Demonstrate the first working iteration and collect feedback.	Demo #1
Nov 10–16, 2026	Apply feedback from Demo #1, expand verified government information, improve interface behavior, and conduct additional system testing.	Report #3 Part 1 – Nov 16
Nov 17–23, 2026	Complete system testing and final documentation. Verify consistency between implementation, requirements, and reports.	Full Report #3 – Nov 23
Nov 24–Dec 3, 2026	Refine the system, resolve remaining defects, improve ministry coverage, and prepare the final demonstration.	Final refinement
Dec 4, 2026	Present the completed SuperAsk.AI system.	Demo #2

The Gantt chart below summarizes the planned development activities and major project milestones from Report #1 through the final demonstration.



Product Ownership and Team Responsibilities

SuperAsk.AI is being developed collaboratively by six team members. Each member has a primary project role based on their technical and organizational strengths, while major design, testing, and implementation decisions are discussed collectively. The Project Lead coordinates work between the different roles and monitors progress against the semester schedule.

Team Member	Role	Work Completed / So Far	Current Responsibilities	Future Responsibilities
Yean Fong	**Project Lead**	Coordinated the project proposal, project scope, team organization, scheduling, and requirements preparation. Contributed to design and implementation documentation.	Coordinate Report #1, requirements, deadlines, communication between team members, and overall project consistency.	Monitor milestones, coordinate system integration and demos, assist testing/documentation, and manage final project delivery.
Phallamonunhh Chun	**AI/ML Engineer**	Contributed to business analysis, AI planning, cybersecurity considerations, and the SuperAsk.AI concept.	Develop and evaluate the RAG pipeline, model integration, retrieval behavior, and AI guardrails.	Improve answer accuracy, retrieval performance, refusal behavior, and bilingual AI handling.
Bunkheang Heng	**Web/API Developer**	Contributed technical development, system architecture planning, full-stack concepts, and database planning.	Develop the web application, APIs, database connections, and integration with the AI system.	Complete integration, optimize backend functions, fix technical defects, and support deployment.
Doungkhe Neang	**Data/Content Curator**	Contributed data organization, content planning, documentation, and source research.	Collect, organize, verify, and categorize official government content used by the system.	Expand ministry coverage, maintain source metadata, and help ensure information remains current and properly cited.

Sereyprophea Sarin	**QA/Testing Lead**	Contributed documentation, communication, interface discussion, and initial testing planning.	Develop test questions and check accuracy, bilingual responses, source citation, and refusal behavior.	Conduct integration and user testing, record defects, collect user feedback, and verify corrections before demonstrations.
Heejae Choi	**UX/UI Designer**	Contributed interface planning, UX/UI concepts, page layout, and prototype development.	Refine user flows, chat interface, responsive layouts, and presentation of answers and citations.	Improve the interface based on user testing and ensure consistency across desktop/mobile views and supported channels.

Management and Coordination

The Project Lead will maintain the overall project schedule and coordinate dependencies between the sub-teams. Development tasks will be reviewed regularly to identify delays or technical problems early. Team members remain primarily responsible for their assigned areas, but implementation and testing will overlap when integration requires cooperation between AI, content, interface, backend, and QA work.

Team Member	Competencies	Assigned Area
Yean Fong	Project management, organizational skills, assertive communication, Python, C++, JavaScript, HTML, CSS, Design	**Management / Integration**
Phallamonunhh Chun	UX/UI, Business Analysis, Cybersecurity, CSS, C++, AI	**AI / Data**
Bunkheang Heng	Technical development, System architecture, Full Stack Development, Database Design, Machine Learning	**Development**

Doungkhe Neang	UX/UI, data curation and management, technical documentation, information organization	**AI / Data**
Heejae Choi	UX/UI, interface planning, C++, JavaScript, HTML, CSS, documentation	**UX / UI**
SereyProphea Sarin	Documentation, communication, UX/UI, Java, JavaScript, HTML, CSS	**Testing / QA**

References

SuperAsk.AI Team AO3. (2026). ***SuperAsk.AI GitHub repository*****.**
https://github.com/Monunhh/Super-ask.git
Used for project source code, version control, and team development.

Open Government Products. (2026). ***AskGov Singapore*****. Singapore Government.**
https://ask.gov.sg/
Used as a reference for how government information can be organized by agency and presented through a searchable question-and-answer interface. AskGov lists government organizations and public-service questions in one place.